

Attendance



Micromouse

Lecture 2: PCB Design

Meeting 10/23

Updates

- Canvas is almost ready!
 - Will be out soon™
- Micromouse website is ready!
 - https://ieee.ics.uci.edu/micromouse/mm_index.html
 - Quick overview



Downloading KiCad 9

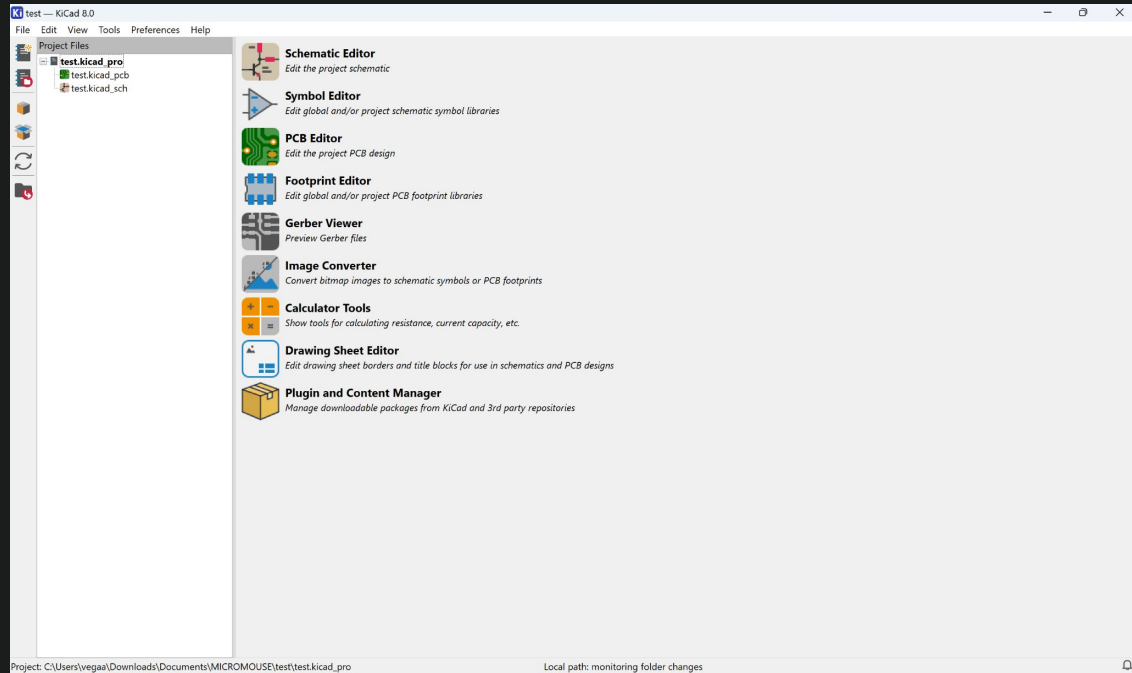
- <https://www.kicad.org/download/>

Why Do We Make PCBs?

- More robust than breadboards
- More versatile than breadboards
 - Custom schematic, layout, soldering, etc.
- More compact than breadboards
 - Can fit smaller SMD components
- Easier to solder than perfboards



Making a New KiCad Project



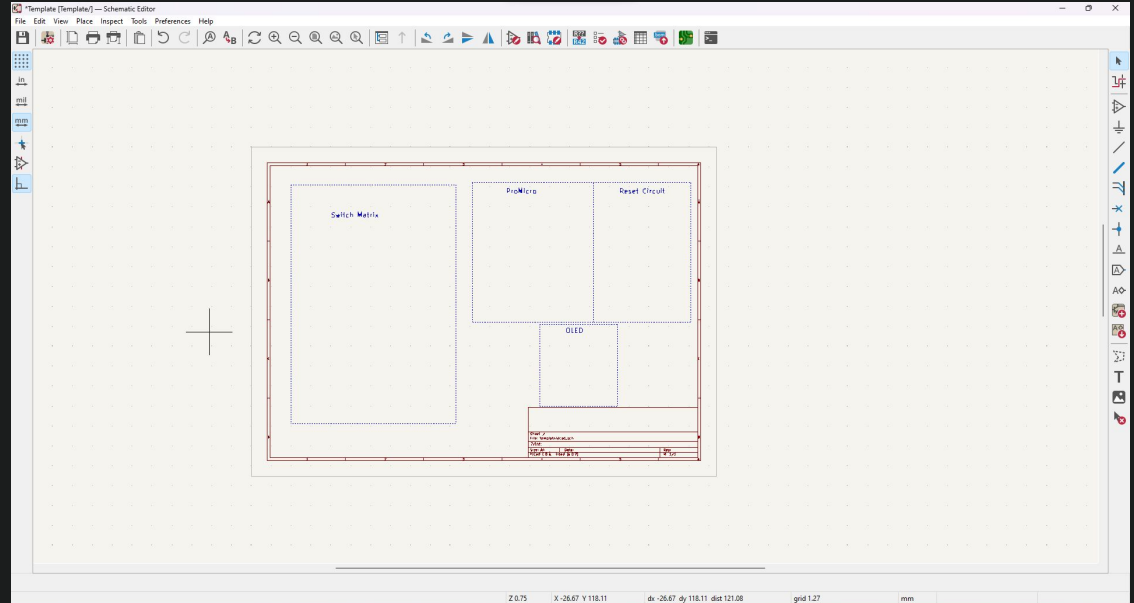
Schematic Layout

Opening the Schematic

- Click the Schematic Editor Button

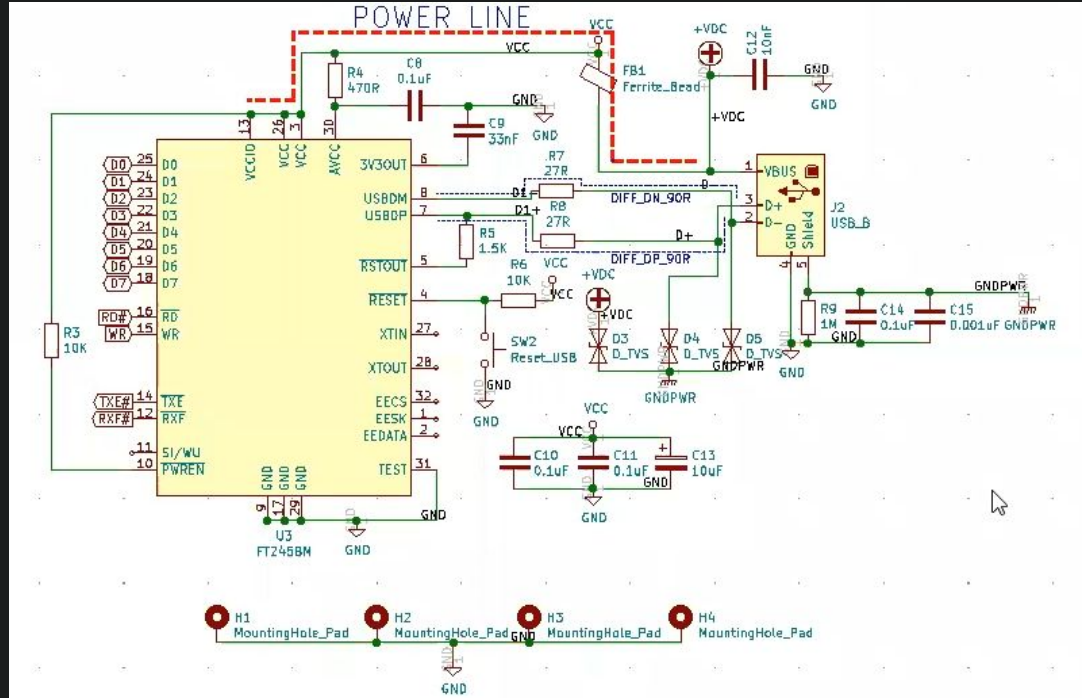


- A window with a canvas should open up →



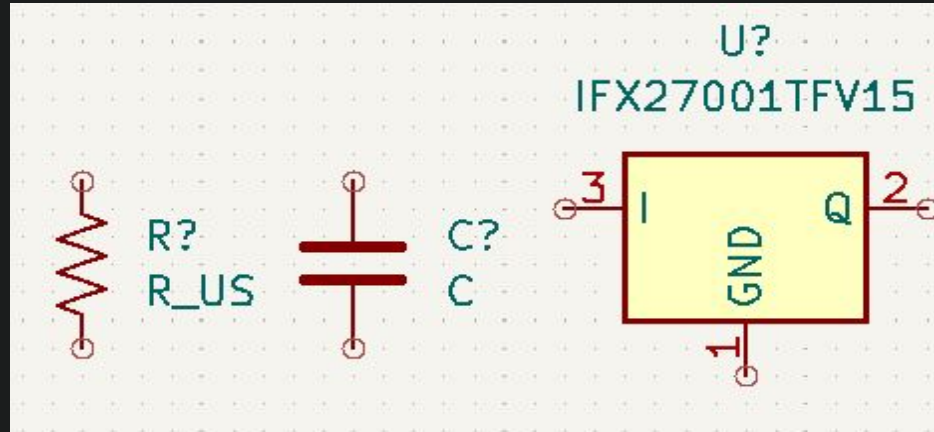
Overview of Developing the Schematic

1. Create schematic by adding symbols
2. Connect wire together
3. Assign footprints to symbols
4. Run ERC

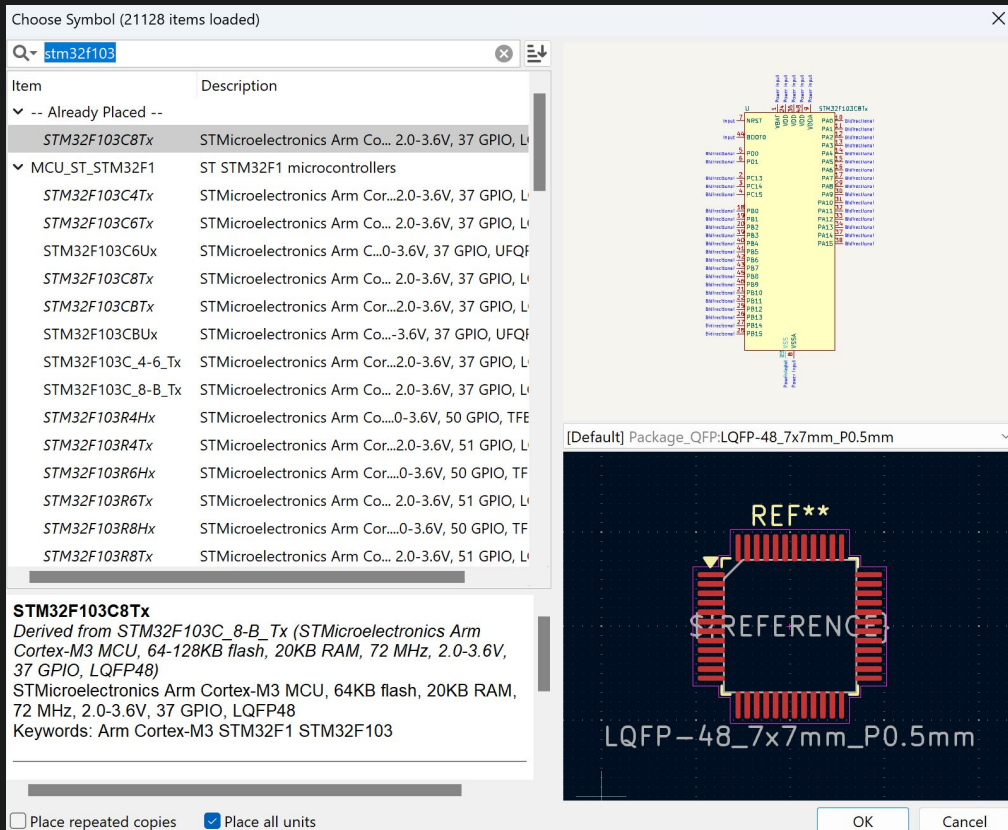
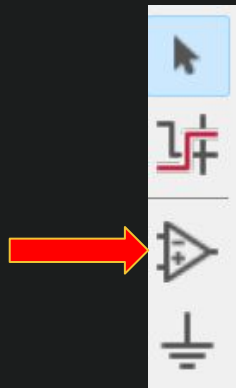


What is a Symbol?

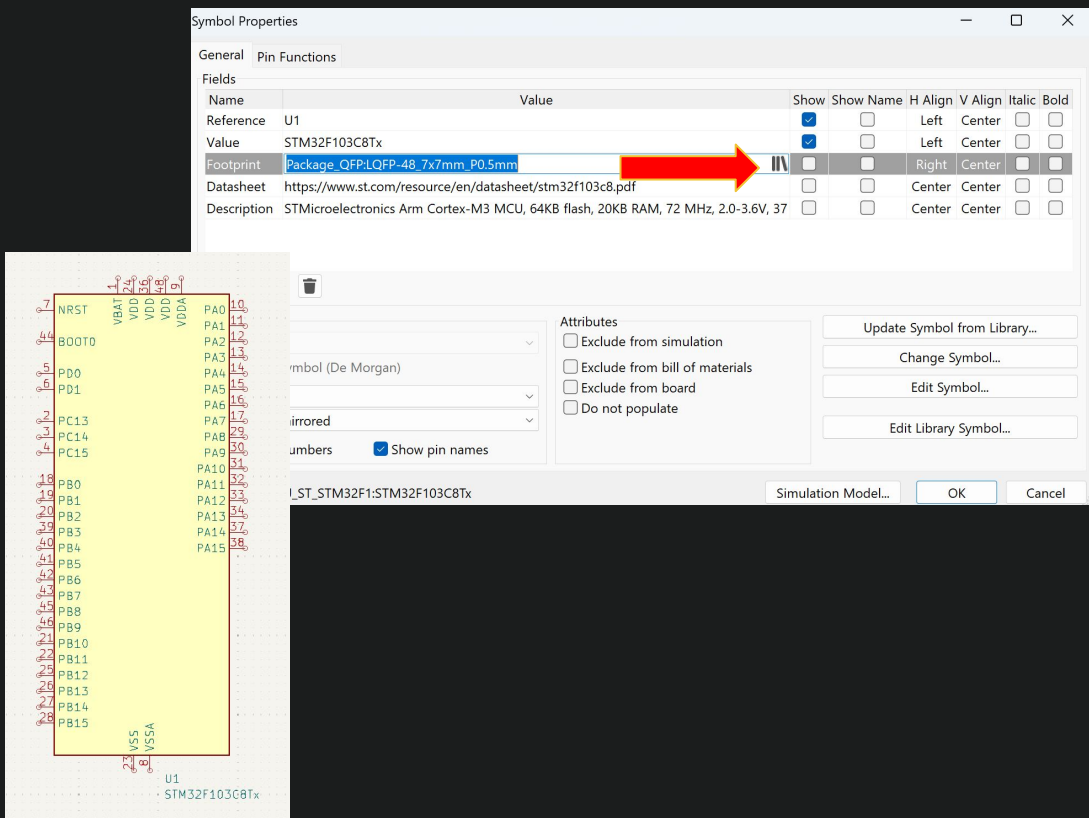
- Simplified representation of parts in a schematic
 - Adding resistors, capacitors, buttons, microcontrollers, etc.
- Connected to each other by wires as a sketch



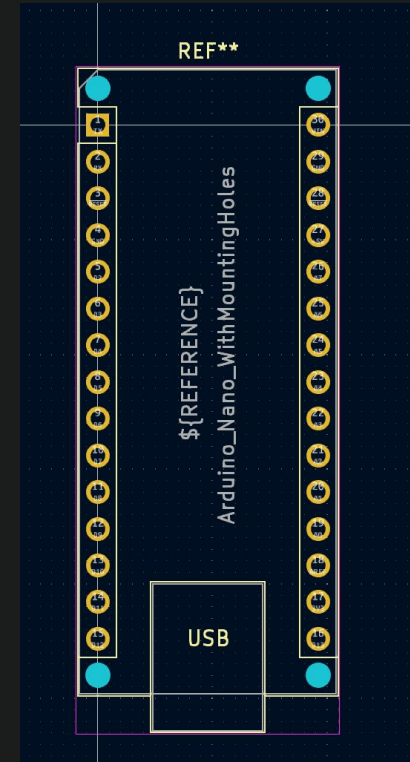
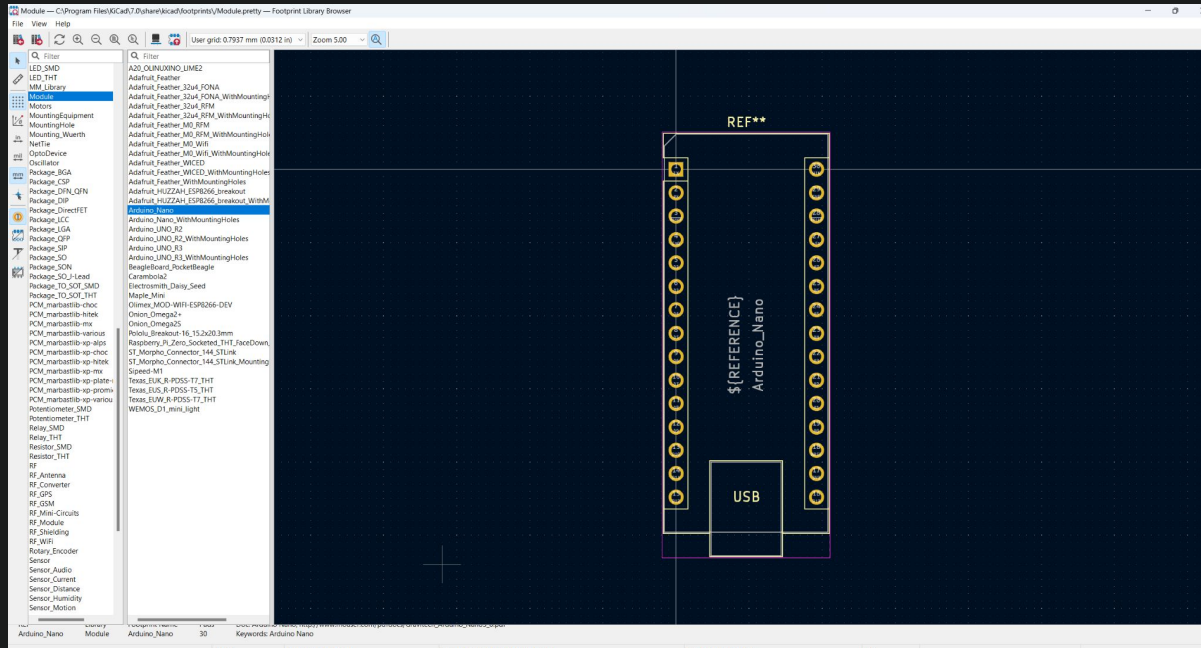
How to Add a Symbol to Schematic



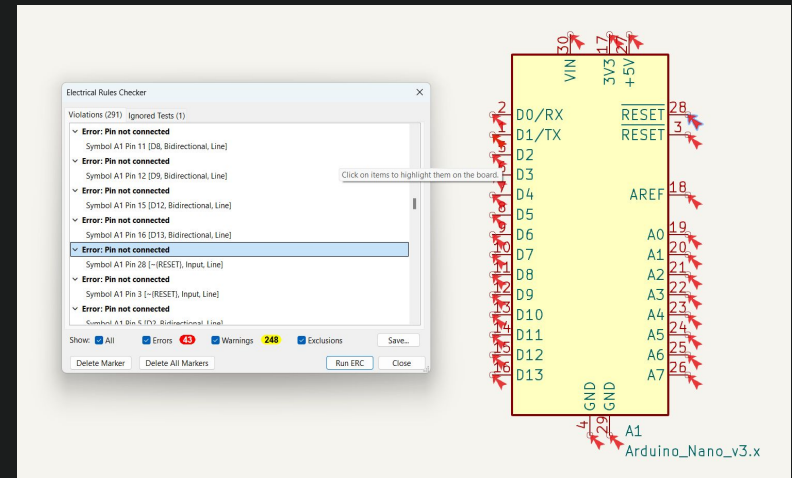
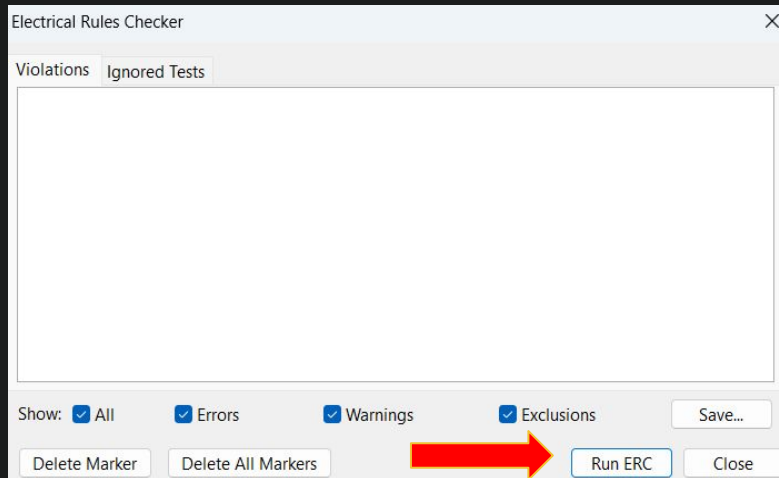
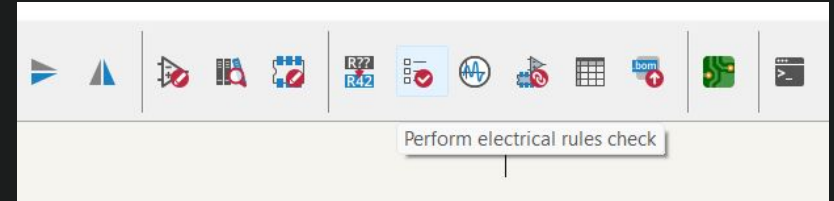
How to Edit Symbol Properties + Assigning Footprints



Assigning Footprints cont.



Running ERC



Shortcuts for Movement

mouse scroll - zoom in and out

CTRL + mouse scroll - move left and right

SHIFT + mouse scroll - move up and down

Left Click - Drag the canvas

Shortcuts for Components

A - add components

V - edit component values

R - rotate the component

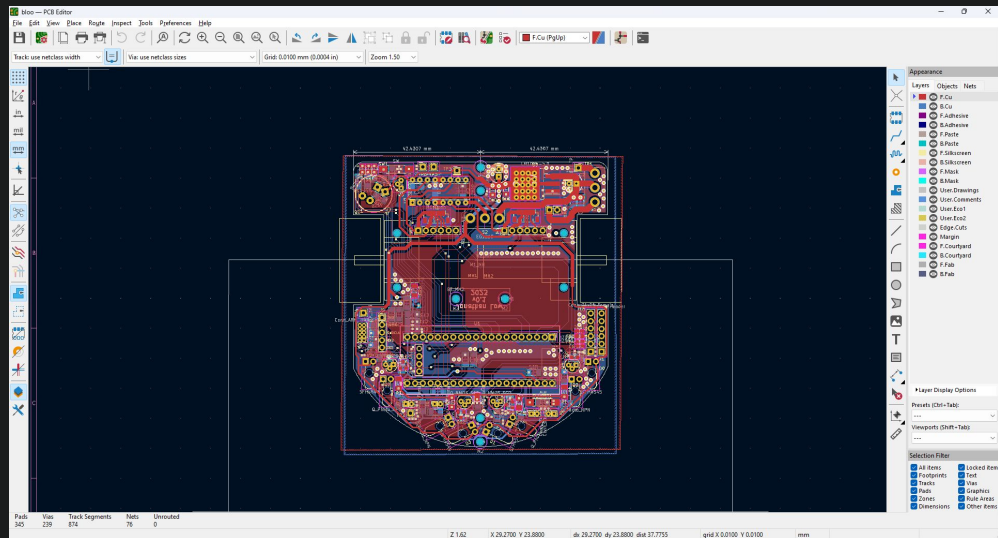
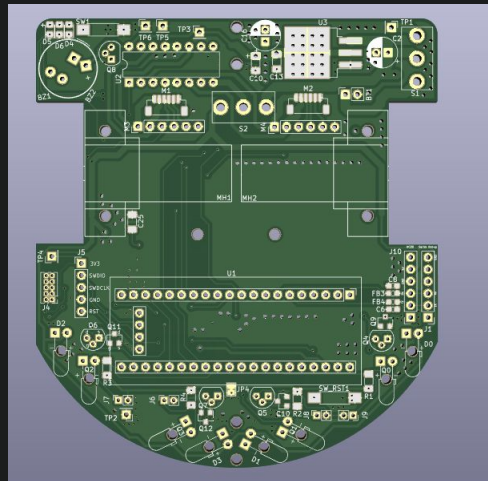
E - edit

Esc - return back to pointer mode

PCB Layout

Overview of Developing PCB Layout

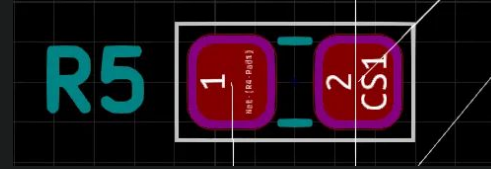
1. Update PCB from Schematic
2. Create board outline
3. Place components
4. Wire traces
5. Run DRC
6. View in 3D Viewer



Frequently Used Terminology

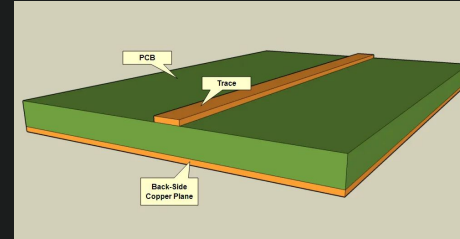
Footprint

Actual shape of a symbol

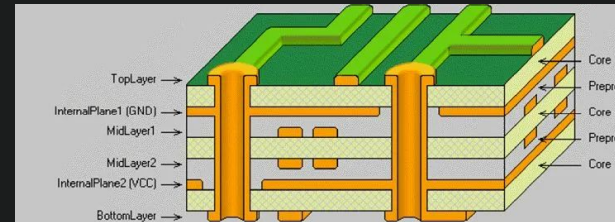


Trace/Track

Copper wire on PCB

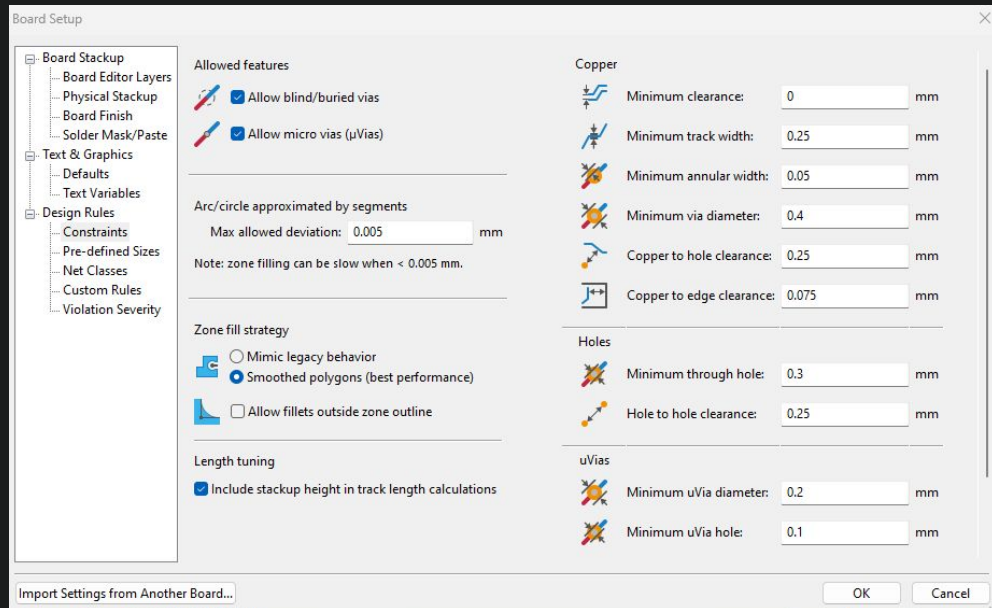
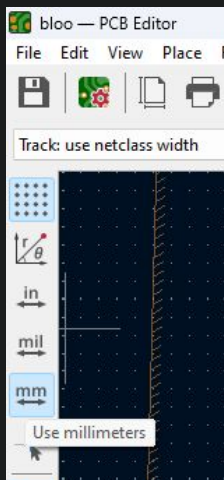
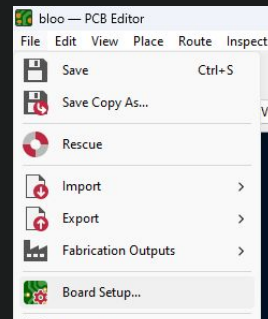


Connects traces on different layers



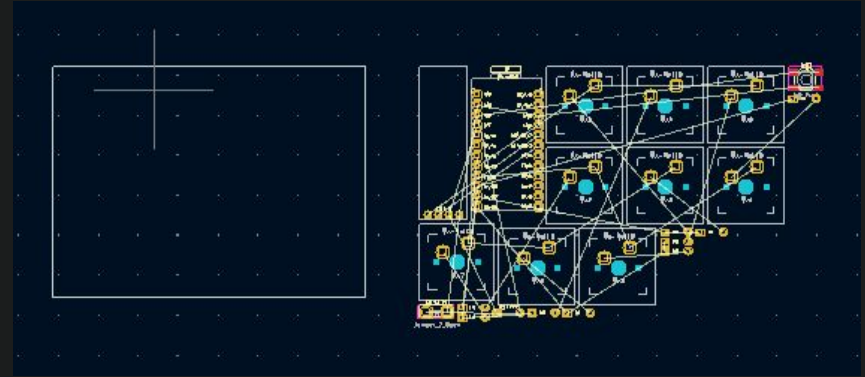
Initial Setup: Check Project Design

- Default trace width of .25mm
 - PCB Editor > File > Board Setup > Design Rules > Constraints
- Set project to be in millimeter

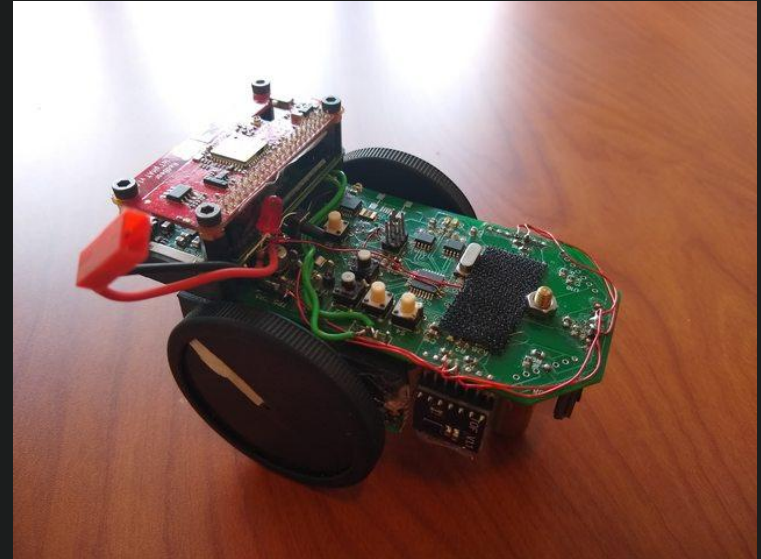


Transfer Footprints to PCB

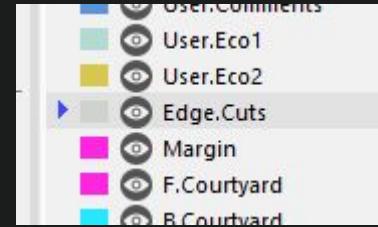
- Click **Tools > Update PCB from Schematic**
- The footprints should open in a new canvas
 - Components will be clustered together
 - Your responsibility to move around footprints on your mouse!



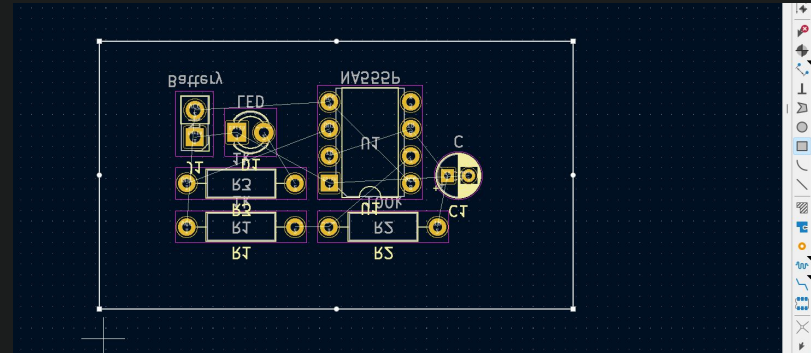
PCB Outline



How to Make PCB Outline

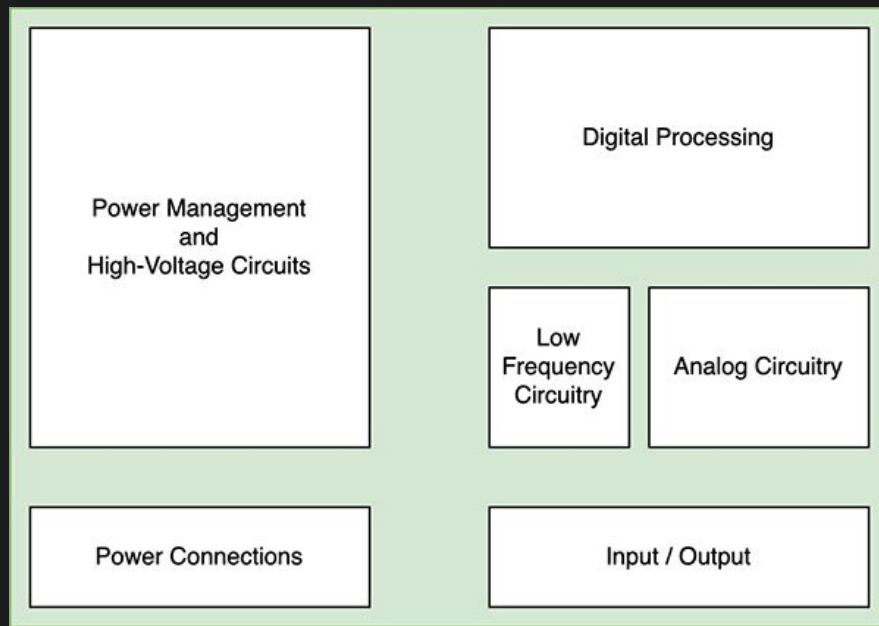


Example:



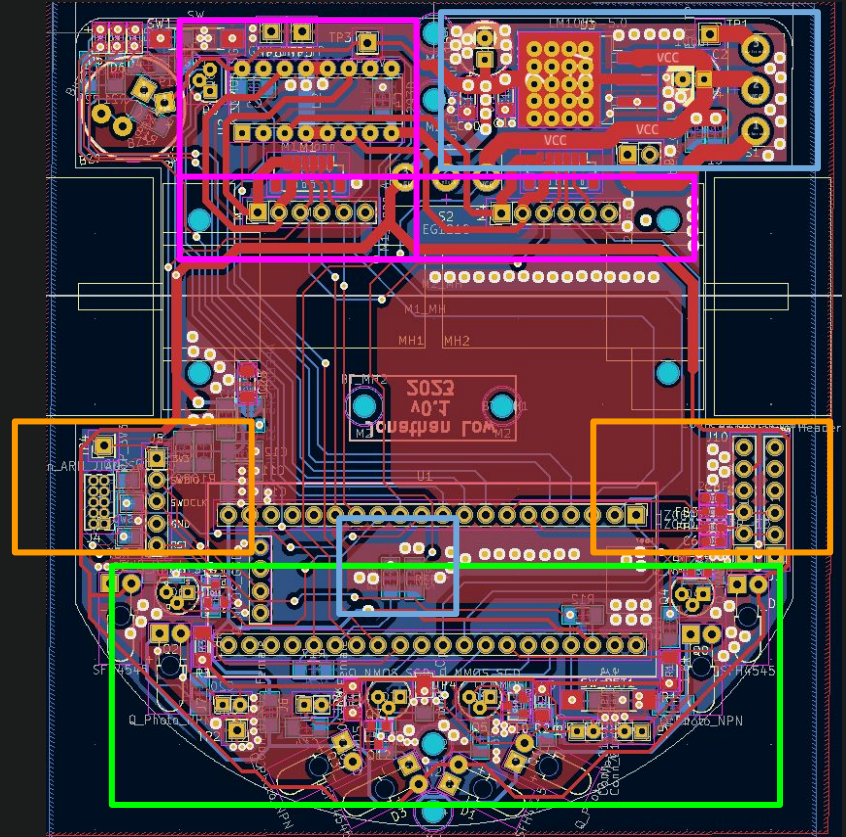
Sub-Circuits/Modularity

- Organize components into subcircuits on PCB
 - Easier to troubleshoot
 - Shorter trace between parts in same subcircuit



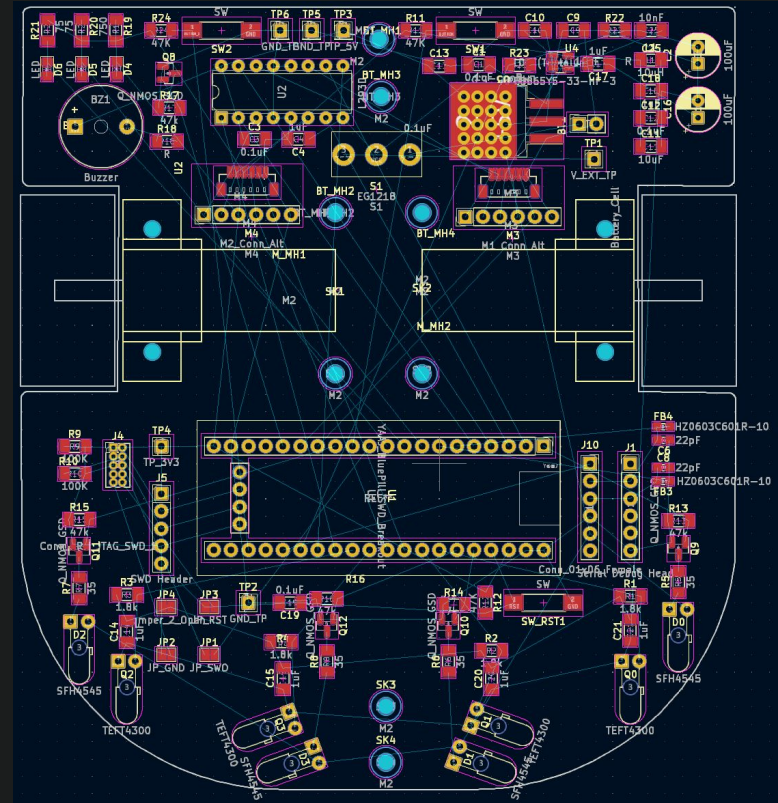
Sub-Circuits Example

- Blue: High voltage - hot!
- Pink: Motor
- Orange: Header pins/buttons/LEDs (things you frequently interface with)
- Green: Sensors (IRs)



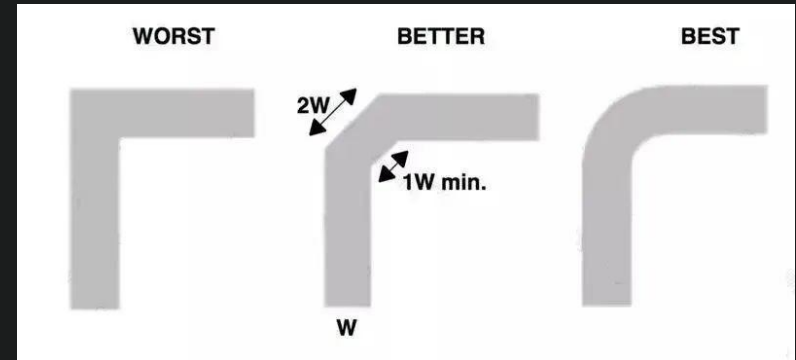
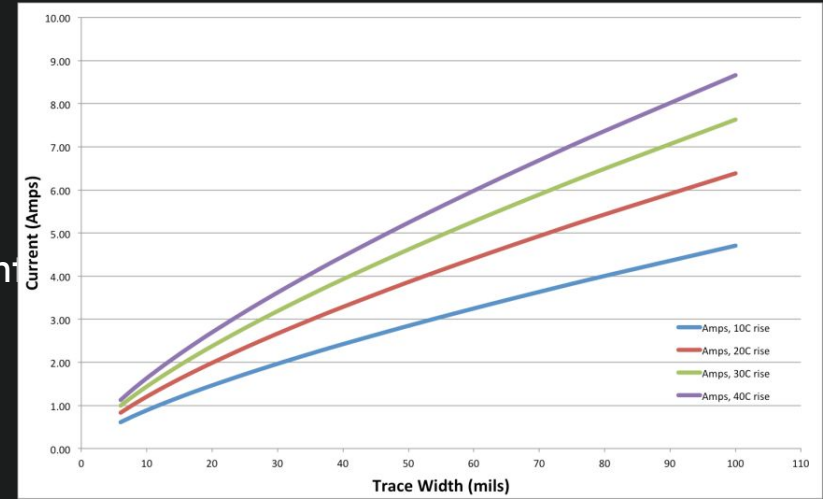
PCB Placement Operation

1. Drag components around and place in subcircuits
2. You will notice faint lines
 - i. These represent what should be connected to each other!



Traces Rule of Thumb

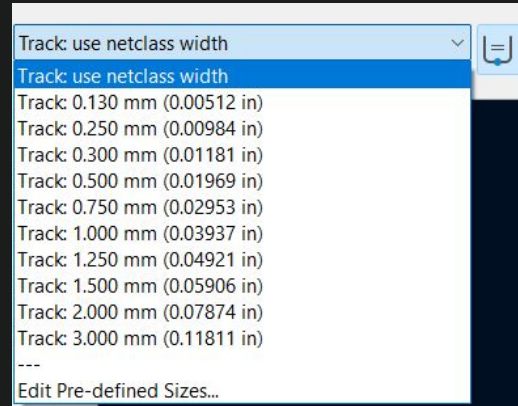
- Connect footprints with traces
 - Think of them as wires between components
- Width matters!
- Equal trace length for pairs
- Shortest possible
- 45° bends
- High speed signals on same layer



Traces Rule of Thumb cont.

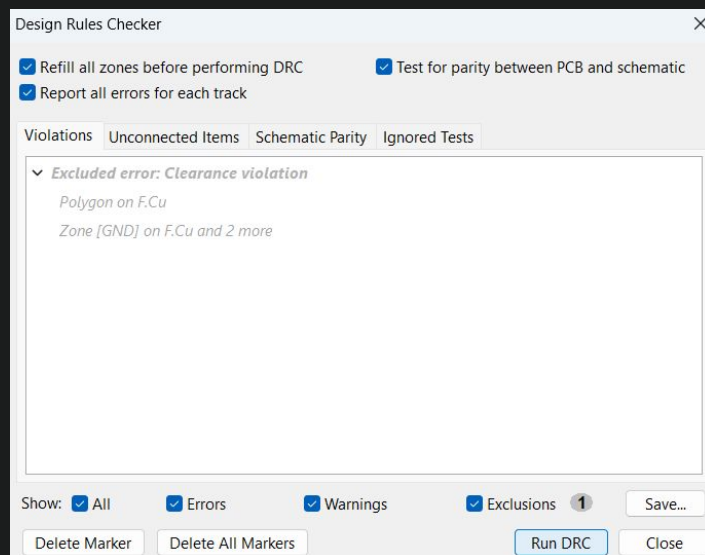
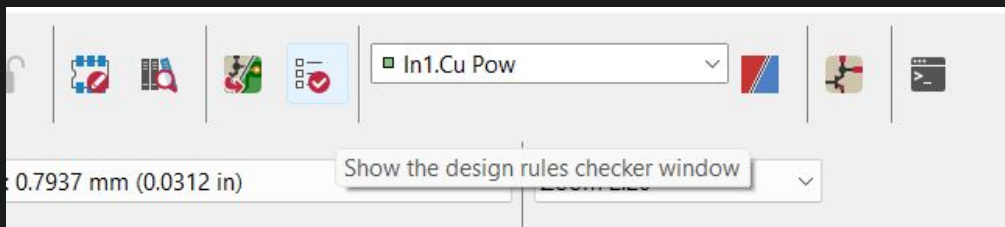
- Power traces should be thicker than others
 - Power traces: 3.3V (.5mm), 5v (.5-1.25mm), VCC (1-1.5mm)
- Traces .25mm apart
- Signal traces all on same layer

Define trace width here (top bar):



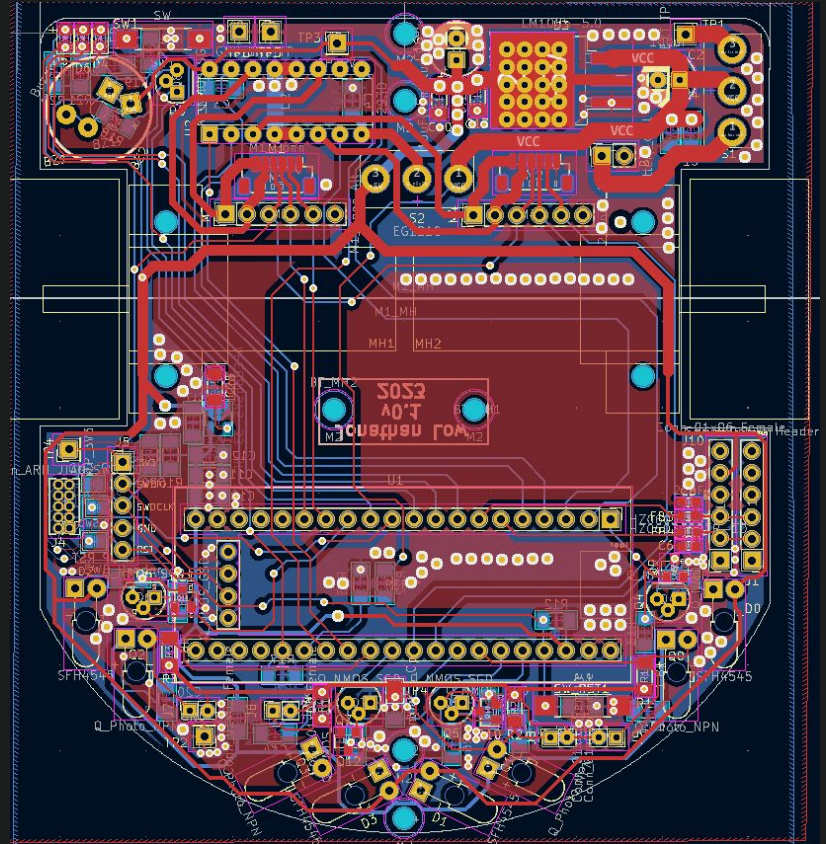
Running/Resolving Common DRC Issues

- Unconnected Items
- Courtyard Overlap
- Silkscreen Overlap

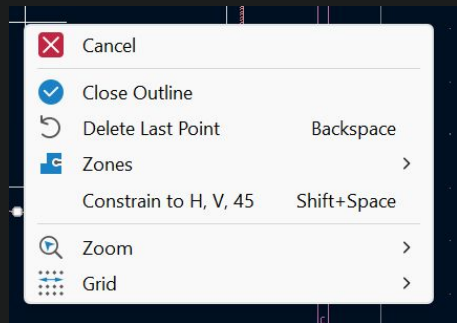
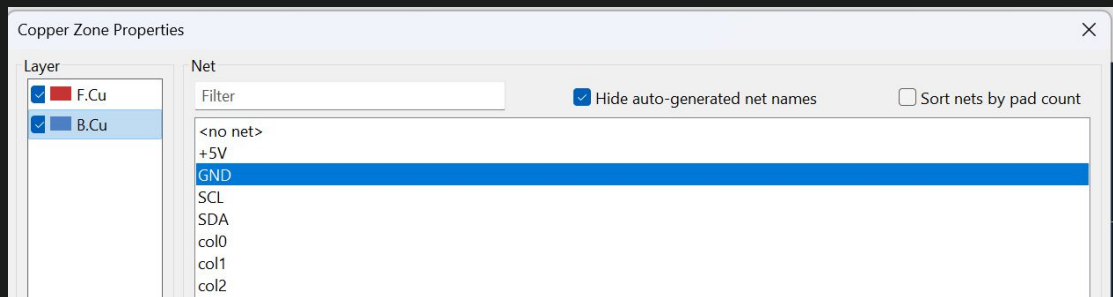


Ground Planes

- Why do they exist?
 - Electromagnetic Interference (EMI)
 - Thermal Issues
 - Easier to ground components
 - Lower impedance
- For your footprints, you don't connect ground nodes with traces - you make a ground plane!



Copper Fill for GND



Shortcuts

R - Rotate Footprint

X - Create a Trace

V - Switches layer, creates Via if you have trace

W/Shift + W - Changes width of trace

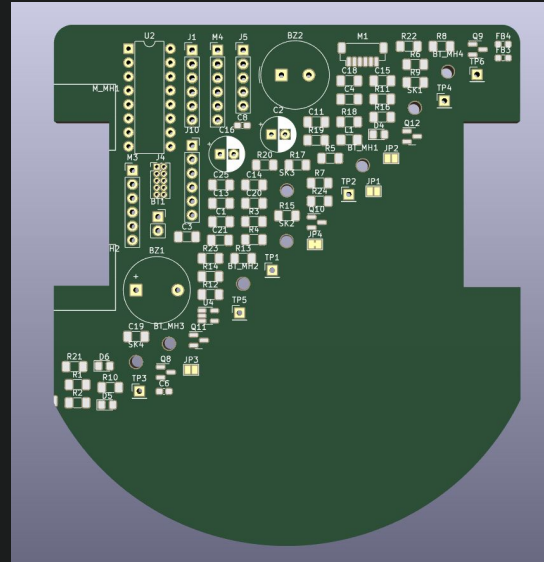
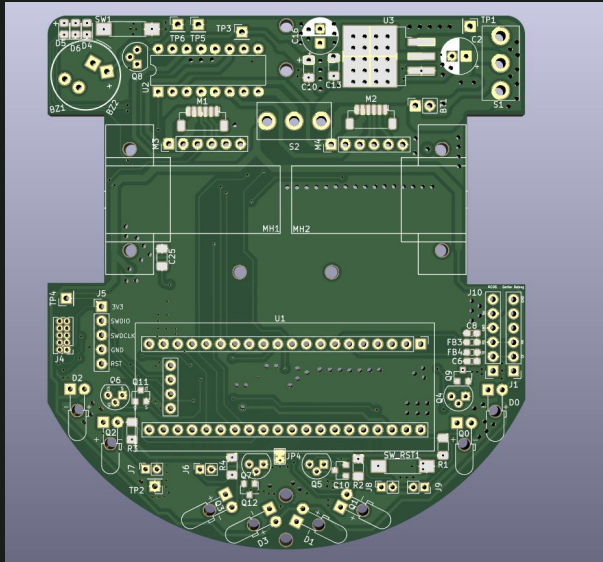
F - Switch footprint to different layer

B - Fill zones

Esc - Back to pointer mode

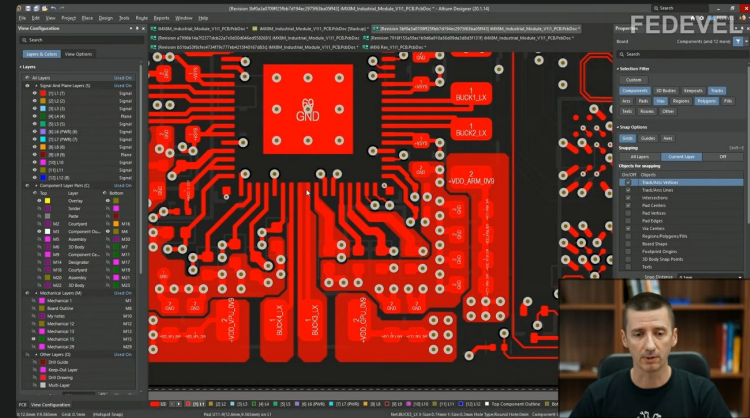
Important Notes

Always make sure to check 3D Viewer! Especially before routing!



Additional Tools

- [Saturn PCB Design](#)
- [PCB Trace Width Calculator](#)
- [Robert Feranec](#)





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PCB Trace Width Calculator

Printed Circuit Board Width Tool

This Javascript web calculator calculates the trace width for printed circuit board conductors for a given current using formulas from IPC-2221 (formerly IPC-D-375).

Inputs:

Current	10	Amps
Thickness	2	oz ² /2

Optional Inputs:

Temperature Rise	10	Deg C
Ambient Temperature	25	Deg C
Trace Length	1	inch

Results for Internal Layers:

Required Trace Width	mil
Resistance	Ohms
Voltage Drop	Volts
Power Loss	Watts

Results for External Layers in Air:

Required Trace Width	mil
Resistance	Ohms
Voltage Drop	Volts
Power Loss	Watts

Notes:

The trace width is calculated as follows:
First, the Area is calculated.

Engineering/Technical

[connection center](#)[SET INSTANT QUOTE](#)[Place PCB & Assembly Order](#)[Capabilities](#)[PCB Materials](#)[FREE PCB Layout Software](#)[FREE PCB File Check](#)[Student Program](#)[Monthly Specials](#)[Download Presentation](#)[DOWNLOAD PRESENTATION](#)

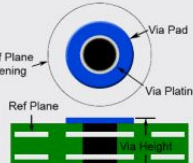
Saturn PCB Design, Inc. - PCB Toolkit [www.saturnpcb.com](#)

[Program Function](#) [Tools](#) [Help](#) [Contact Saturn PCB Design, Inc.](#)

[Conductor Spacing](#) [Conductor Impedance](#) [Conversion Data](#) [Planar Inductors](#) [PDN Impedance](#) [Thermal](#)

[Via Properties](#) [Conductor Properties](#) [Signal Properties](#) [Differential Pairs](#) [Padstack Calculator](#) [Mechanical Information](#)

Via Characteristics



Ref Plane Opening

Via Pad

Via Plating

Via Hole Diameter	10 mils
Internal Pad Diameter	20 mils
Ref Plane Opening Diam	40 mils
Via Height	62 mils
Via Plating Thickness	1 mils

Via Capacitance	0.4021 pF
Via Inductance	1.3262 nH
Via Impedance	57.429 Ohms

Via DC Resistance	0.0013 Ohms
Resonant Frequency	6891.661 MHz
Step Response	25.4032 ps

Power Dissipation	0.002 Watts
Conductor Cross Section	34.5575 Sq.mils
Via Current	1.1113 Amps

Options

Base Copper Weight	Imperial
Units	Imperial
Substrate Options	FR-4 STD
Material Selection	Er 4.6 Tg (C) 130
Plating Thickness	Temp Rise (C) 10
Temp Rise (C)	Temp in (F) 36.0
Plane Thickness	Ambient Temp (C) 22
Temp in (F) 71.6	
Layer Set	2 Layer
Multi Layer	
Print	Solver

Information

Total Copper Thickness N/A

Conductor Temperature Temp in (C) = 42.0 Temp in (F) = 107.6



Next Steps

- Watch this video! Full walkthrough of PCB design
 - <https://youtu.be/3akdNExVPvI?si=NeWbXYzk99Jn97Uu>
- We will give you the complete schematic! It is your responsibility to develop the PCB layout

- Week 7 (11/14 - ICS259): KiCAD/PCB Design Lecture/Setup
- Week 8 (11/21 - EH2430): Kit Mouse Routing Initial Review & Team Finalization (due **Friday, 11/22/24, 11:59pm**)
- Week 9: Thanksgiving Break (no meeting)
- Week 10 (12/5 - EH 2430): Kit Mouse Routing Final In-Person Review (final routing due **Friday, 12/13/24, 11:59pm**)
- Week 11: Finals (good luck!)

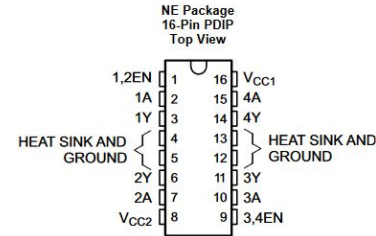
Best Practices

- Trace width for power vs signal
- Following spec sheet practices
- Capacitors near components
- Same trace length for pairs of traces (ie: TX, RX)
- Traces as short as possible
- Organization of sub-circuits
- Ground Planes
- 45° bends
- Vias in Grid to avoid Hotspots
- High Speed Signal in 1 layer

Spec Sheet

- Part sizes
- Pin Configuration
- Characteristics
- Operation Modes
- Much more!

5 Pin Configuration and Functions



Pin Functions

PIN		TYPE	DESCRIPTION
NAME	NO.		
1,2EN	1	I	Enable driver channels 1 and 2 (active high input)
<1,4>A	2, 7, 10, 15	I	Driver inputs, noninverting
<1,4>Y	3, 6, 11, 14	O	Driver outputs
3,4EN	9	I	Enable driver channels 3 and 4 (active high input)
GROUND	4, 5, 12, 13	—	Device ground and heat sink pin. Connect to printed-circuit-board ground plane with multiple solid vias
V _{CC1}	16	—	5-V supply for internal logic translation
V _{CC2}	8	—	Power VCC for drivers 4.5 V to 36 V

10 Power Supply Recommendations

V_{CC1} is 5 V ± 0.5 V and V_{CC2} can be same supply as V_{CC1} or a higher voltage supply with peak voltage up to 36 V. Bypass capacitors of 0.1 μF or greater should be used at V_{CC1} and V_{CC2} pins. There are no power up or power down supply sequence order requirements.

Properly heatsinking the L293 when driving high-current is critical to design. The R_{thj-amp} of the L293 can be reduced by soldering the GND pins to a suitable copper area of the printed circuit board or to an external heat sink.

Figure 14 shows the maximum package power P_{TOT} and the θ_{JA} as a function of the side of two equal square copper areas having a thickness of 35 μm (see Figure 14). In addition, an external heat sink can be used (see Figure 12).

During soldering, the pin temperature must not exceed 260°C, and the soldering time must not exceed 12 seconds.

The external heatsink or printed circuit copper area must be connected to electrical ground.